ID: Assessor(s):

Artifact Expectations and Structured Feedback

The following expectations for the assignment were met:

☒ Design Dossier submitted through SharePoint ☒ Total file size for Design Dossier is less than 5GB ☒ Responsibility Tracker included ☒ Procurement & Budget Tracker included

Design Dossier Axes of Evaluation N = in range; U(L) = Developing (Low); D(L, M, H) = Developed (Low, Middle, High); Q = High Quality

		Development Level							
Component	**Elements**	**N**	**U**	**UH**	**DL**	**DM**	**DH**	**Q**	**Comments**
Prototyping Outcomes	Prototype Overview	☐	☐	☐	☐	☐	☐	☐	
	Specification	☐	☐	☐	☐	☐	☐	☐	
	Prototype System Features & Functionality	☐	☐	☐	☐	☐	☐	☐	
	Coherence across the Design Chain	☐	☐	☐	☐	☐	☐	☐	
Prototype System Design Process	Prototype System Design Process (`04.1_PrototypeDesign`)	☐	☐	☐	☐	☐	☐	☐	
	Prototype Subsystem Implementation Process (`04.2_Build`)	☐	☐	☐	☐	☐	☐	☐	
	Systems Integration & Debugging Process (`04.3_Integration`)	☐	☐	☐	☐	☐	☐	☐	
	Verification Protocol + Results (`04.4_Verification`)	☐	☐	☐	☐	☐	☐	☐	
Teamwork Artifacts (Teamwork Process & Outcomes)	Team Setup: Team Charter	☐	☐	☐	☐	☐	☐	☐	
	Team Progress Tracking: Project Schedule	☐	☐	☐	☐	☐	☐	☐	
	Team Dynamics: `05.0_EngineeringNotebook`	☐	☐	☐	☐	☐	☐	☐	

ID: Assessor(s):

Holistic Score

Quality Score X.X / 10

Adjusted Score N/A / 10

Holistic Comments

Feedback Guide

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Prototype Overview	Prototype Overview is not present, or lacks sufficient detail to meaningfully summarize prototyping activities undertaken during Phase 3; OR Prototype Overview does not incorporate or contain any references to additional artifacts from elsewhere in the Dossier (i.e., figures, verification results, etc.)	Prototype Overview is present, and helps to outline the Prototype System design, implementation, integration and verification processes which were undertaken during Phase 3; however, a lack of detail (or concision), poor organization, or missing elements limits its utility as an overall summary of prototyping work. The as-built Prototype System is presented in the Prototype Overview, but more information is needed to fully understand how it works. Prototype Overview may incorporate or reference other Dossier artifacts, but such references do not always enhance understanding of the design process or outcomes.	Prototype Overview is present, and concisely summarizes the Prototype System design, implementation, integration and verification processes which were undertaken during Phase 3. The as-built Prototype System is clearly presented with enough detail to fully understand how it works. Key design decisions and implementation challenges faced are highlighted. References to other Dossier artifacts enhance understanding of the design process and outcomes. The Prototype Overview provides a roadmap to understand the work completed by the team throughout Phase 3.	Developed + the Prototype Overview offers a concise, precise summary of both the process taken to arrive at the as-built Prototype System and the Prototype System itself. Use of figures, tables and other organizational elements as necessary enhances readability. Emphasis is given to key design decisions both during initial design activities and implementation/integration, which are presented with appropriate justification; challenges in implementation and system integration (if encountered) are discussed with reference to appropriate evidence, and possible solutions for future design work are offered.
Specification	No Specification is provided, or the desired functions, performance or behaviour of the Prototype System is not clear from the provided Specification.	A Specification is provided and outlines key behaviour for the Prototype System, however it may be unclear how the Specification relates to the NGOs for the Design Concept; OR The Specification provided is not focused on the Prototype System such that many elements do not appear to be verifiable using the actual Prototype System to be realized within the scope of the course.	A Specification is present in the Design Dossier, which outlines the key function(s) and performance of the Prototype System, distinct from but related to that for the Design Concept. Specification elements clearly connect relevant NGOs (in particular, Objectives) for the Design Concept to the desired behaviours of the Prototype System. Specification elements are focused on the actual Prototype System to be realized, and are generally verifiable within the scope of the course.	"Developed" + the Specification provided both clearly outlines the desired functions/performance of the Prototype System (and how it should achieve them) relative to those for the Design Concept, while enabling useful design work to take place as the Prototype System is created. Many possible Prototype Systems exist which could meet this Specification, and each of them would meet the goals of the designers in exploring the Design Concept further.

ID: Assessor(s):

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Prototype System Features & Functionality	System is not functional or fails to meet key elements of the developed Specification, with no explanation/exploration. OR Insufficient evidence is provided in the Design Dossier to demonstrate key Prototype System functionality. OR Prototype System or design/implementation process violates Course or Design Process Constraints.	Video and other evidence of verification results are provided to show system in action. Prototype System and its subsystems meet some but not all key elements of the developed Specification, with no explanation/exploration of shortcomings.	Video and other evidence of verification results are provided to show the Prototype System in action, and convincingly prove functionality. The Prototype System meets the developed Specification; OR Prototype System does not meet some parts of the Specification, but there is clear explanation/discussion of how to remedy any shortcomings, and/or an interpretation of what these shortcomings demonstrate about the Design Concept.	"Developed" + construction of the Prototype System is high quality (e.g. smooth mechanical action, robust joints; well commented and organized code; clean circuits with colour-coded wires, repeatable, reliable and accurate sensing, etc.). Complexity is incorporated into the Prototype System in a way that enhances functionality and demonstrates the team's progress towards relevant Key Competencies for the course. Note that choosing to solder circuits does **not** impact construction quality in this row.
Coherence across the Design Chain	It is unclear from the provided artifacts, considered along with the Specification and Prototype Overview, how the Prototype System connects to the (potentially revised) Design Concept, Engineering Requirements, and overall opportunity context.	Evidence of connection(s) between design chain elements (Prototype System, Design Concept, Engineering Requirements, and opportunity context) is present, but may be shallow in places. The Prototype System may feel disconnected from the overall design context, or it is unclear what is gained through the prototyping activities (i.e., what has the team learned about the Design Concept from constructing the Prototype System, even though the Prototype System is not equivalent to the overall Design Concept?).	Evidence of connection(s) between design chain elements is present when Dossier artifacts are considered along with the Specification and Prototype Overview, and demonstrates strong coherence (i.e., logical links, supported by appropriate engineering evidence) between engineering decisions made in each space on the Design Chain. <u>If</u> any framing updates were made from Phase 2, these are captured through revisions/annotations to the Design Concept Overview, Engineering Requirements spreadsheet, and/or Prototype Concept (as necessary).	"Developed" + links between design chain elements are clear and explicit in Dossier artifacts; it is immediately clear to the reader how the Prototype System supports investigation of the Design Concept through the Specification, and how the Design Concept is connected to the opportunity context through the (potentially revised) Engineering Requirements.
Prototype System Design Process	The design process (creation of plans for the Prototype System prior to, and potentially revised during, implementation) is not demonstrated through included artifacts.	Design process is demonstrated through included artifacts; however, a lack of organization of artifacts or synthesis through the Prototype Overview leads to a limited understanding of the design process. Documentation may be insufficient to enable understanding of key design decisions which allow the Prototype System to achieve function(s) and/or performance outlined in the Specification.	Design process (distinct from but complementary to the implementation process) is clearly demonstrated through included artifacts and synthesis through the Prototype Overview. The process is reasonable and complete, and key design decisions to meet the developed Specification are highlighted when process artifacts are considered along with the Prototype Overview.	"Developed" + the design process is thorough, methodical, and justified. Key design decisions are supported with research, use of reference designs, or preliminary (lower-fidelity) exploration or implementation, which have been captured through artifacts in the Design Dossier.

ID: Assessor(s):

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Prototype Subsystem Implementation Process	Implementation (build) processes for individual Prototype subsystems are not clear based on included artifacts.	Implementation process(es) for individual subsystems are demonstrated, but present artifacts may lack detail or present an incomplete picture, which leaves the reader with a limited understanding of the build process when process artifacts are considered along with the Prototype Overview.	Implementation process(es) for individual subsystems are demonstrated with sufficient detail to support any detailed design decisions (i.e., changes to the subsystem design) made during implementation when process artifacts are considered along with the Prototype Overview.	"Developed" + the implementation process is methodical, and includes justification for approach and tool/component choices throughout. Documentation is thorough, and the team demonstrates resilience to implementation challenges.
Systems Integration & Debugging Process	System integration process for the full Prototype System is not clear based on provided artifacts; OR No integration took place.	System integration process is demonstrated (and Prototype System is clearly integrated), but steps taken and/or obstacles faced may not be clear from process artifacts when considered along with the Prototype Overview. OR It is clear that considerations for systems integration were only made retroactively, rather than proactively during design.	Integration process for the Prototype System is demonstrated through the provided artifacts when considered along with the Prototype Overview, with sufficient detail for a reader to understand any obstacles faced, and the associated debugging process. Some consideration for Systems Integration (forethought) was incorporated into the design and implementation of subsystems.	"Developed" + systems integration is methodical and well-documented. Preliminary testing of interfaces between subsystems at various stages of integration demonstrates successes and challenges during debugging.
Verification Protocol & Results	There is no evidence of systematic verification of the Prototype System against the Specification, either planned or conducted; OR The Verification Protocol is missing significant information (i.e., a Verification Procedure is absent for multiple Specification elements); OR Verification Procedures are not clearly linked to a corresponding Specification element; OR Fewer than 2 specification elements are verified through Testing (as opposed to Demonstration, Analysis or Investigation).	For most Specification Elements, there is a Verification Procedure provided which captures the performance of the Prototype System against each element of the provided Specification; however, procedures (especially for "Demonstrate" and "Test" elements) are vague such that it is unclear how verification was conducted; AND/OR There is some evidence of Verification which generally follows any present Procedures, but verification may appear incomplete or invalid.	For each Specification Element, there is a Verification Procedure provided which captures the performance of the Prototype System with respect to that element. Enough detail is provided (particularly for "Demonstrate" and "Test" elements) to clearly outline how each Verification Procedure was conducted; AND Verification is performed using the Prototype System, and results are clearly recorded though video, photographs, or other means appropriate to the Verification Procedure.	"Developed" + approach to "Test" elements is systematic and methodical. Test results include response to edge cases and show robustness (or lack thereof!) of the Prototype System. Insights gained from verification enhance the team's understanding of the Design Concept as well as the Prototype System.

ID: Assessor(s):

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Team Setup: Team Charter	There is no evidence/documentation of team members' values, backgrounds, team identity, or the teamwork strategy through a complete Team Charter.	Limited documentation of the team, their values, backgrounds, and team identity is presented in the Team Charter, which makes the document feel tangential rather than integral to the teamwork processes practiced during Phase 3. OR There is no evidence that the Team Charter has been updated to reflect teamwork review(s) conducted by the team after Phase 2.	Documentation of the team, their values, backgrounds, and team identity through the Team Charter is detailed. The Team Charter is integral to teamwork processes practiced in Phase 3, as demonstrated by evidence of team practices that align with the established protocols throughout Phase 3, or analysis of how team practices aligned (or not) with the established protocol. At least one review of the Team Charter is documented.	"Developed" + review(s) of the teamwork process have clearly brought the Team Setup into closer alignment with identified team values.
Team Progress Tracking: Project Schedule	There is no plan presented for prototyping work in Phase 3, nominal or actual.	A Project Schedule for tasks related to prototyping work in Phase 3 is presented; however, the plan may feel disconnected from the team, their values, preferred workstyles, or other constraints outlined in the Team Charter; alternatively, the plan may represent only a nominal plan for Phase 3, without any documentation of the actual timeline, delays experienced, etc.	The Project Schedule for prototyping work in Phase 3 appears consistent with the team's values, workstyles, and/or other constraints as laid out in the Team Charter. Both a nominal and actual plan for the semester is presented, with discrepancies between the two noted and explained.	"Developed" + the Project Schedule clearly shows the nominal and actual timelines realized for prototyping, and includes discussion of key blockers, challenges, or other circumstances which impacted the timeline, along with suggestions for how to avoid these discrepancies in the future (possibly through reference(s) to other Dossier artifacts).
Team Dynamics: 05.0_EngineeringNotebook	The Engineering Notebook doesn't exist; OR It fails to include records of the team's work beyond meeting minutes or other basic records of team progress.	There is little evidence of implementation of the Engineering Notebook as a primary record of the team's activities (beyond cataloging team meetings); artefacts included as part of the Notebook may not contain sufficient detail to be of later use to the team, or may not capture important design work completed by the team throughout Phase 3.	The Engineering Notebook effectively serves as a primary record of the team's work during Phase 3, by including a relatively complete record of both team meetings (Meeting Minutes) and project work (Work Logs) completed throughout. Notebook entries have sufficient detail for an observer to understand what took place (and when). Overall, the Notebook appears to be used effectively by the team as a record of work taking place.	"Developed" + The Engineering Notebook is integrated into teamwork processes, and is clearly integral to driving the team's design work forward.